

Question 1

T1

Jack rolled a water bottle from one edge of his desk to the other. The desk measures 60 cm from edge to edge. The water bottle has a diameter of 7 cm. Determine the angle that the water bottle rotated, in degrees.

Solution

$$s = \theta r$$

$$\theta = \frac{s}{r}$$

$$\theta = \frac{60}{3.5}$$

$$\theta = 17.142\,857\dots$$

1 mark for substitution

$$\theta = (17.142\,857\dots) \left(\frac{180^\circ}{\pi} \right)$$

1 mark for conversion

$$\theta = 982.213\,363\dots^\circ$$

$$\theta = 982.213^\circ$$

2 marks

Question 2

P3

There are 15 dogs and 12 cats in an animal shelter. Determine the number of ways that three dogs and two cats can be selected if Scout, one of the dogs, must be selected.

Solution

$${}_1C_1 \cdot {}_{14}C_2 \cdot {}_{12}C_2$$

6006

1 mark for ${}_{14}C_2$

$\frac{1}{2}$ mark for ${}_{12}C_2$

$\frac{1}{2}$ mark for product of combinations

2 marks

Note:

${}_1C_1$ does not need to be shown.

Question 3

R10

Emily wants to save money to buy a car. She invests \$180 per month at an annual interest rate of 4.5%, compounded monthly.

Determine, algebraically, the number of monthly investments she will need to make to obtain at least \$15 000. Express the final answer as a whole number.

Use the formula: $FV = \frac{R[(1+i)^n - 1]}{i}$

where FV = the future value

R = the investment amount each period

$$i = \left[\frac{\text{the annual interest rate (as a decimal)}}{\text{the number of compounding periods per year}} \right]$$

n = the number of investments

Solution

$$15\,000 = \frac{180 \left[\left(1 + \frac{0.045}{12} \right)^n - 1 \right]}{\frac{0.045}{12}}$$

½ mark for substitution

$$15\,000 = \frac{180 \left[(1 + 0.00375)^n - 1 \right]}{0.00375}$$

$$0.3125 = 1.00375^n - 1$$

$$1.3125 = 1.00375^n$$

½ mark for simplification

$$\log(1.3125) = \log(1.00375)^n$$

½ mark for applying logarithms

$$\log(1.3125) = n \log(1.00375)$$

1 mark for power law

$$n = \frac{\log(1.3125)}{\log(1.00375)}$$

$$n = 72.651\,539\dots$$

½ mark for evaluating quotient of logarithms

∴ 73 monthly investments are needed

3 marks

Question 4

P4

Determine and simplify the 4th term in the binomial expansion of $\left(3x - \frac{2}{x^2}\right)^6$.

Solution

$$t_4 = {}_6C_3 (3x)^3 \left(-\frac{2}{x^2}\right)^3$$

2 marks (1 mark for ${}_6C_3$; $\frac{1}{2}$ mark for each consistent factor)

$$= 20(27x^3) \left(\frac{-8}{x^6}\right)$$

$$= \frac{-4320}{x^3}$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponent)

3 marks

Question 5

T5

Solve, algebraically, over the interval $[0, 2\pi]$.

$$4\cos^2 x - 3\cos x - 1 = 0$$

Solution

$$4\cos^2 x - 3\cos x - 1 = 0$$

$$(4\cos x + 1)(\cos x - 1) = 0$$

$$\cos x = -\frac{1}{4} \qquad \cos x = 1 \qquad 1 \text{ mark for solving for } \cos x \text{ (}\frac{1}{2} \text{ mark for each branch)}$$

$$x_r = 1.318 \ 116 \dots$$

$$x = 1.823, 4.460 \qquad x = 0, 2\pi \qquad 2 \text{ marks for solving for } x \text{ (}\frac{1}{2} \text{ mark for each value)}$$

3 marks

Question 6

R14

State an equation for a rational function, $g(x)$, whose graph has a vertical asymptote at $x = 7$.

Solution

$$g(x) = \frac{1}{x-7}$$

1 mark

Note:

Other equations are possible.

Question 7

R13

State an equation of a radical function, $f(x)$, with a domain of $x \leq 0$ and a range of $y \geq 1$.

Solution

$f(x) = \sqrt{-x} + 1$ 1 mark for a radical function with a domain of $x \leq 0$

1 mark for a radical function with a range of $y \geq 1$

2 marks

Note:

Other equations are possible.

Question 8

T6

Prove the identity for all permissible values of x .

$$\frac{\cos x + \sin^2 x \sec x}{\sin x} = \sec x \csc x$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\cos x + \sin^2 x \sec x}{\sin x}$	$\sec x \csc x$
$\frac{\cos x + \frac{\sin^2 x}{\cos x}}{\sin x}$	
$\frac{\cos^2 x + \sin^2 x}{\cos x \sin x}$	
$\frac{1}{\cos x \sin x}$	
$\frac{1}{\cos x} \cdot \frac{1}{\sin x}$	
$\sec x \csc x$	

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 8

T6

Method 2

Left-Hand Side	Right-Hand Side
$\frac{\cos x + \sin^2 x \sec x}{\sin x}$	$\sec x \csc x$
$\frac{\cos x}{\sin x} + \frac{\sin^2 x \sec x}{\cancel{\sin x}}$	
$\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}$	
$\frac{\cos^2 x + \sin^2 x}{\sin x \cos x}$	
$\frac{1}{\sin x \cos x}$	
$\sec x \csc x$	

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 8

T6

Method 3

Left-Hand Side	Right-Hand Side
$\frac{\cos x + \sin^2 x \sec x}{\sin x}$	$\sec x \csc x$
$\cos x + \frac{1 - \cos^2 x}{\cos x}$	
$\frac{\cos^2 x + 1 - \cos^2 x}{\cos x}$	
$\frac{1}{\cos x \sin x}$	
$\sec x \csc x$	

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

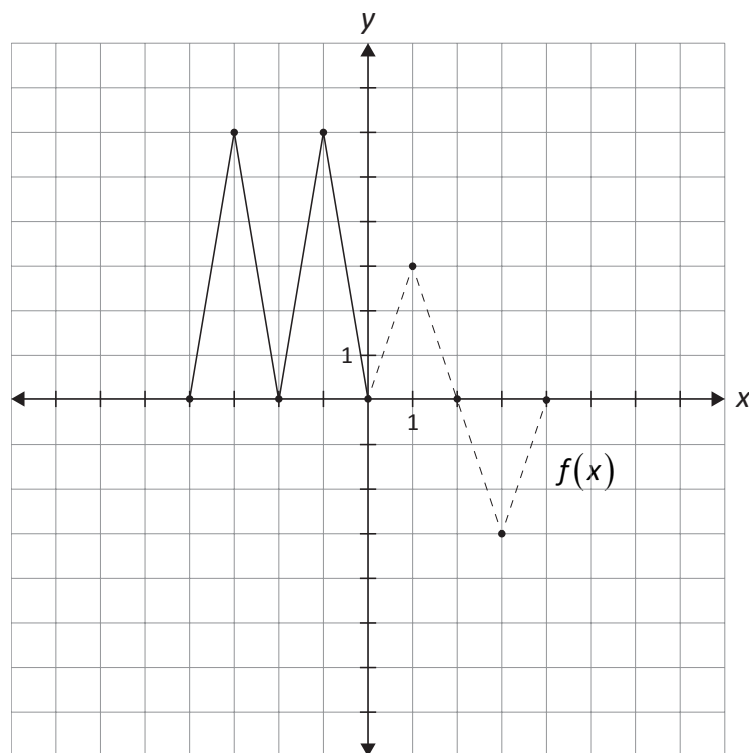
3 marks

Question 9

R1, R3, R5

Given the graph of, $y = f(x)$, sketch the graph of $y = 2|f(-x)|$.

Solution



1 mark for horizontal reflection
1 mark for absolute value
1 mark for vertical stretch

3 marks

Question 10

R10

Solve, algebraically.

$$\log(x-1) + \log(x+2) = 1$$

Solution

$$\log[(x-1)(x+2)] = 1 \quad 1 \text{ mark for product law}$$

$$(x-1)(x+2) = 10^1 \quad 1 \text{ mark for exponential form}$$

$$x^2 + x - 2 = 10$$

$$x^2 + x - 12 = 0$$

$$(x+4)(x-3) = 0$$

$$x = \cancel{-4} \quad x = 3$$

$\frac{1}{2}$ mark for the permissible value of x
 $\frac{1}{2}$ mark for showing the rejection of the extraneous root

3 marks

Question 11

P4

Justify that there are only two negative terms in the expansion of $(-2x + 5y)^4$.

Solution

$$t_1 = {}_4C_0(-2x)^4(5y)^0 \rightarrow +$$

$$t_2 = {}_4C_1(-2x)^3(5y)^1 \rightarrow -$$

$$t_3 = {}_4C_2(-2x)^2(5y)^2 \rightarrow +$$

$$t_4 = {}_4C_3(-2x)^1(5y)^3 \rightarrow -$$

$$t_5 = {}_4C_4(-2x)^0(5y)^4 \rightarrow +$$

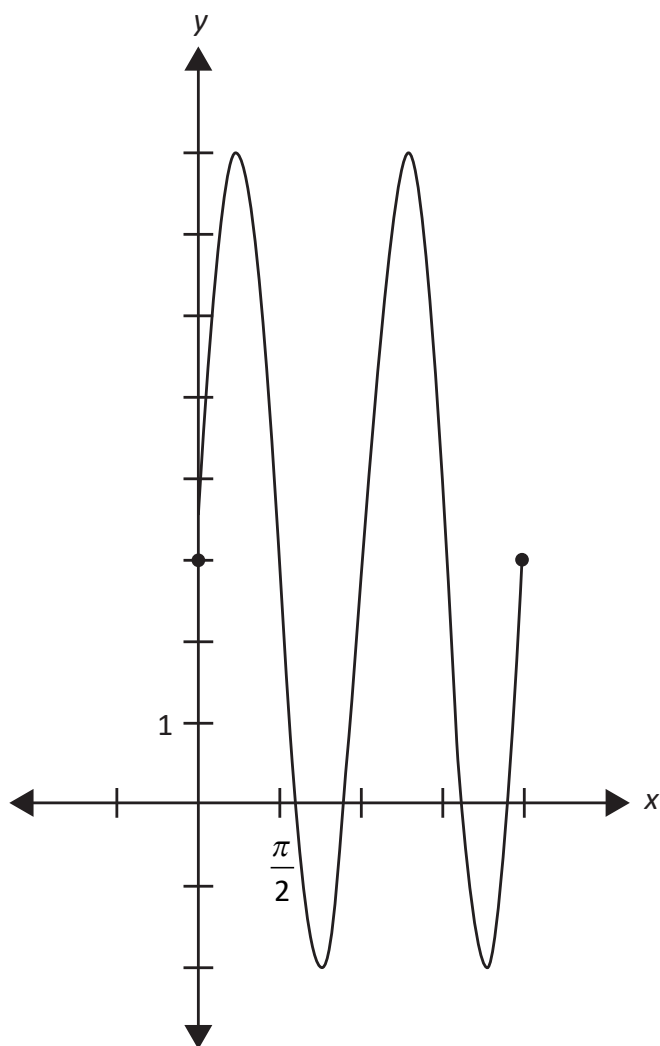
$\therefore$ The term is negative only when $(-2x)$ has an odd exponent, which occurs twice.

1 mark

Question 12

T5

The graph of $y = 5\sin(2x) + 3$ below can be used to solve the equation $0 = 5\sin(2x) + 3$.
State how many solutions there are to the equation $0 = 5\sin(2x) + 3$ over the interval $[0, 2\pi]$.



Solution

There are four solutions.

1 mark

Question 13

R1

Given the functions, $f(x) = x^2 - 1$ and $g(x) = x + 1$,

a) state the equation of $h(x) = \frac{f(x)}{g(x)}$.

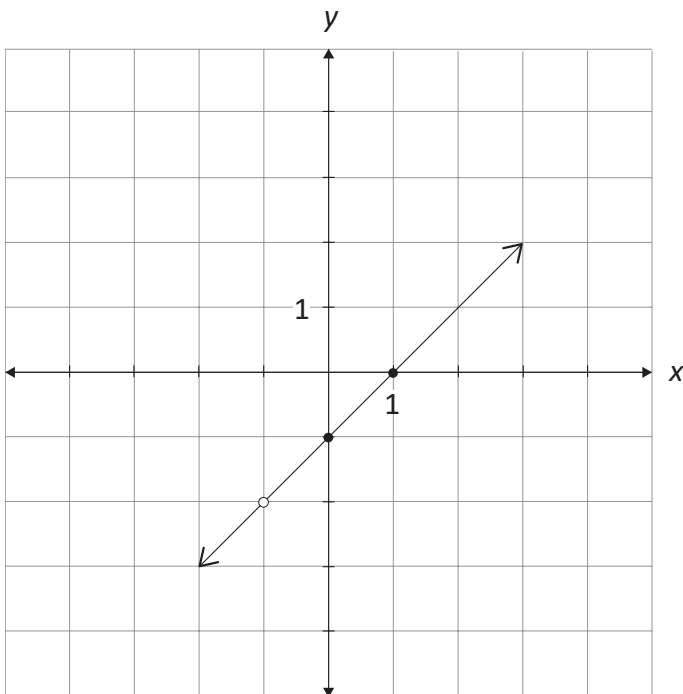
b) sketch the graph of $h(x)$.

Solution

$$\begin{aligned} \text{a) } h(x) &= \frac{x^2 - 1}{x + 1} \\ &= \frac{(x+1)(x-1)}{(x+1)} \\ &= x - 1, x \neq -1 \end{aligned}$$

1 mark

b)



1 mark for shape of graph
consistent with a)

1 mark for point of discontinuity
(hole) at $x = -1$

2 marks

Note:

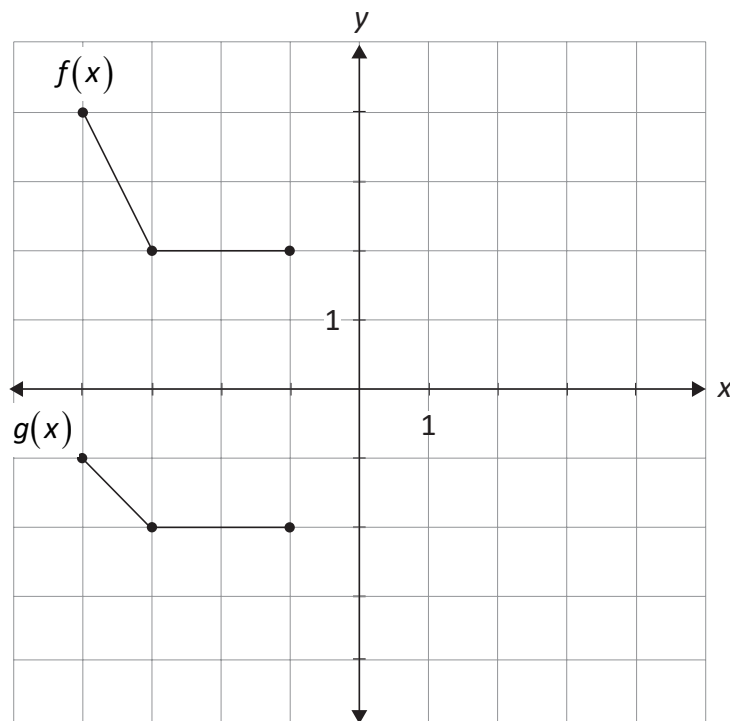
Deduct a maximum of 1 mark for the concept error of not restricting domain.

Deduct ½ mark for procedural error (not stating domain of simplified function in part a)
if graph shows the correct domain.

Question 14

R2, R3

State an equation for $g(x)$, in terms of $f(x)$.



Solution

$$g(x) = \frac{1}{2}f(x) - 3$$

1 mark for vertical compression

1 mark for vertical translation

2 marks

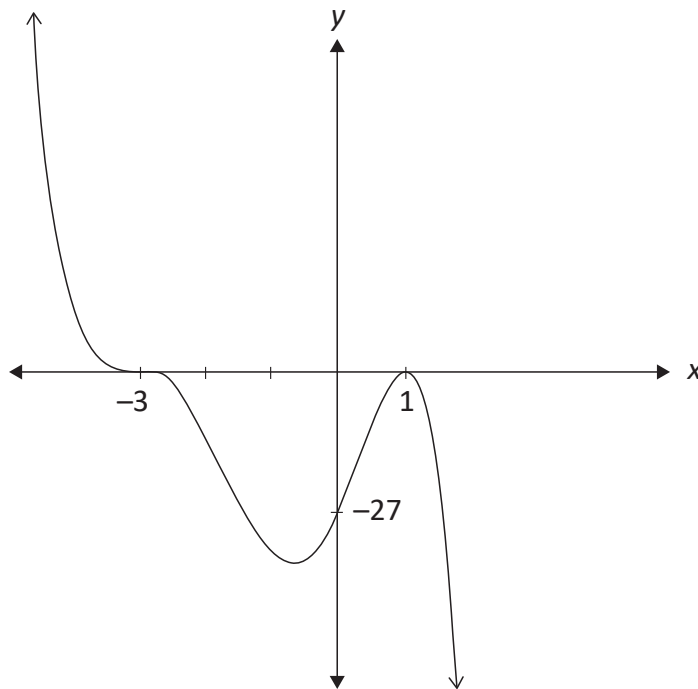
Question 15

R12

Sketch the graph of a polynomial function, $p(x)$, with the following characteristics:

- degree 5
- leading coefficient of -1
- a zero at -3 , with a multiplicity of 3
- a zero at 1 , with a multiplicity of 2

Solution



1 mark for x-intercept at -3
with a multiplicity of 3

1 mark for x-intercept at 1
with a multiplicity of 2

$\frac{1}{2}$ mark for end behaviour

$\frac{1}{2}$ mark for y-intercept

3 marks

Question 16

T6

Identify the expression that is equivalent to $\tan\theta(\csc^2\theta - 1)$.

a. $\cot\theta$

b. $\tan\theta$

c. $\sec\theta$

d. $\sin\theta$

Question 17

P2

Identify the expression that represents the number of ways five books can be arranged on a shelf if two of them must be together.

a. $5!2!$

b. $5! - 2!$

c. $4!2!$

d. $3!2!$

Question 18

R10

Identify the value of x in the equation, $\log_8 x = -\frac{1}{3}$.

a. -2

b. $-\frac{1}{2}$

c. $\frac{1}{2}$

d. 2

Question 19**R1**

Given the functions, $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x-1}$, identify the domain of $f(g(x))$.

- a. $(-\infty, \infty)$
- b. $(-\infty, 1) \cup (1, \infty)$
- c. $[1, \infty)$

d. $(1, \infty)$

Question 20**R14**

Given $f(x) = \frac{3(x+2)}{5(x+2)(x-2)}$, identify the equation of the horizontal asymptote.

- a. $y = -2$

b. $y = 0$

c. $y = \frac{3}{5}$

- d. $y = 2$

Question 21**T4**

Identify the range of the sinusoidal function, $f(x) = 2\sin x + 3$.

- a. $\{y \mid y \in \mathbb{R}\}$
- b. $\{y \mid 2 \leq y \leq 4, y \in \mathbb{R}\}$
- c. $\{y \mid -2 \leq y \leq 2, y \in \mathbb{R}\}$

d. $\{y \mid 1 \leq y \leq 5, y \in \mathbb{R}\}$

Question 22**R11**

Given that $(x-3)$ is a factor of the polynomial $P(x)$, identify which of the following is true.

a. $P(3) = 0$

b. $P(0) = 3$

c. $P(0) = -3$

d. $P(-3) = 0$

Question 23**T1**

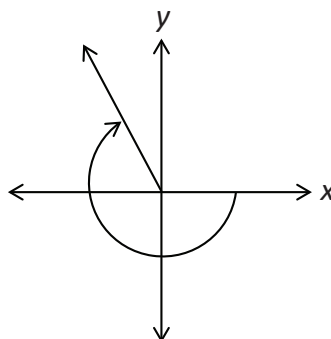
Identify the value that best represents the angle in standard position.

a. -4.4 radians

b. -2 radians

c. 1.3 radians

d. 1.9 radians

**Question 24****R8**

Identify the expression that is equivalent to $\log_a(x+2) + 3\log_a(x)$.

a. $\log_a(3(x+2))$

b. $\log_a(x^3(x+2))$

c. $\log_a\left(\frac{(x+2)}{x^3}\right)$

d. $\log_a x + \log_a 2 + \log_a x^3$

Question 25

T3

Evaluate.

$$\tan^3\left(\frac{3\pi}{4}\right) + \csc\left(\frac{-4\pi}{3}\right) \cos\left(\frac{13\pi}{6}\right)$$

Solution

$$(-1)^3 + \left(\frac{2}{\sqrt{3}}\right)\left(\frac{\sqrt{3}}{2}\right)$$

1 mark for $\tan\left(\frac{3\pi}{4}\right)$ (½ mark for value, ½ mark for quadrant)

$$-1 + 1$$

$$0$$

1 mark for $\csc\left(\frac{-4\pi}{3}\right)$ (½ mark for value, ½ mark for quadrant)

1 mark for $\cos\left(\frac{13\pi}{6}\right)$ (½ mark for value, ½ mark for quadrant)

3 marks

Question 26

R8

If $a = \log 4$ and $b = \log 3$, express $\log 36$ in terms of a and b .

Solution

$$\begin{aligned}\log 36 &= \log 4 + \log 3^2 \\ &= \log 4 + 2\log 3 \\ &= a + 2b\end{aligned}$$

1 mark for product law

1 mark for power law

2 marks

Question 27

T6

Justify that $\tan 2x$ has a non-permissible value at $x = \frac{\pi}{4}$.

Solution

Method 1

$$\tan 2x$$

$$\text{Let } x = \frac{\pi}{4},$$

$$\tan\left(2 \cdot \frac{\pi}{4}\right) \quad 1 \text{ mark for substitution}$$

$$\tan \frac{\pi}{2}$$

$$\frac{1}{0} \text{ which is undefined} \quad 1 \text{ mark for justification}$$

$$\therefore x \neq \frac{\pi}{4}$$

2 marks

Method 2

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x} \quad 1 \text{ mark for correct substitution of appropriate identity}$$

$$1 - \tan^2 x \neq 0$$

$$\tan^2 x \neq 1$$

$$\tan x \neq \pm 1$$

$$\therefore x \neq \frac{\pi}{4} \quad 1 \text{ mark for justification}$$

2 marks

Method 3

$$\tan 2x = \frac{\sin 2x}{\cos 2x}$$

$$\cos 2x \neq 0$$

$$\text{Let } x = \frac{\pi}{4},$$

$$\cos\left(2 \cdot \frac{\pi}{4}\right)$$

1 mark for substitution

$$\cos\left(\frac{\pi}{2}\right) = 0$$

1 mark for justification

$$\therefore x \neq \frac{\pi}{4}$$

2 marks

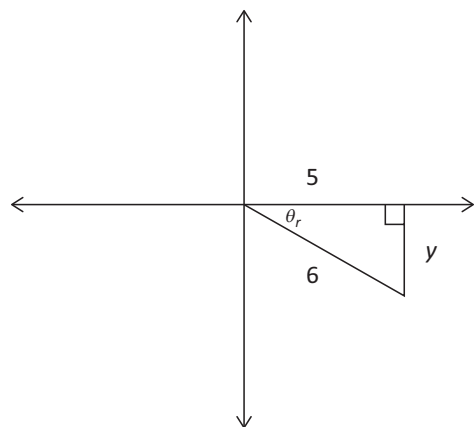
Question 28

T3

Determine the exact value of $\tan\theta$, if $\sec\theta = \frac{6}{5}$ and θ terminates in quadrant IV.

Solution

Method 1



$$\cos\theta = \frac{5}{6}$$

$$5^2 + y^2 = 6^2$$

$$y^2 = 36 - 25$$

$$y^2 = 11$$

$$y = \pm\sqrt{11}$$

$$\tan\theta = \frac{-\sqrt{11}}{5}$$

1 mark for value of y

1 mark for $\tan\theta$ (½ mark for quadrant, ½ mark for value)

2 marks

Note:

Accept any of the following values for y : $y = \pm\sqrt{11}$, $y = \sqrt{11}$, $y = -\sqrt{11}$.

Question 28

T3

Method 2

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$\tan^2 \theta = \left(\frac{6}{5}\right)^2 - 1$$

1 mark for substitution into appropriate identity

$$\tan^2 \theta = \frac{36}{25} - 1$$

$$\tan^2 \theta = \frac{11}{25}$$

$$\tan \theta = \frac{\pm\sqrt{11}}{5}$$

$$\therefore \tan \theta = \frac{-\sqrt{11}}{5} \text{ in quadrant IV}$$

1 mark for $\tan \theta$ (½ mark for quadrant, ½ mark for value)

2 marks

Question 29

R14

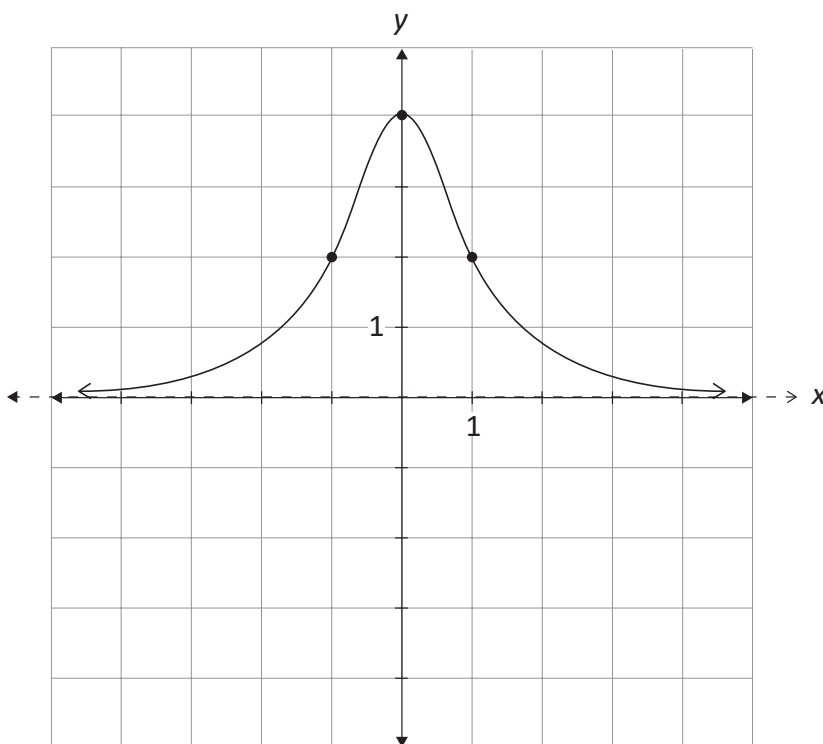
Given the function, $g(x) = \frac{4}{x^2 + 1}$,

a) sketch the graph of $g(x)$.

b) state the range of the graph of $g(x)$.

Solution

a)



1 mark for asymptotic behaviour approaching $y = 0$

1 mark for shape
($\frac{1}{2}$ mark for graph left of $x = 0$;
 $\frac{1}{2}$ mark for graph right of $x = 0$)

2 marks

b) Range: $\{y | 0 < y \leq 4, y \in \mathbb{R}\}$

or

Range: $(0, 4]$

1 mark for range consistent with a)

1 mark

Question 30

R6

Given $y = x^2 - 4$,

a) determine the equation of the inverse.

b) state a restriction on the domain of $y = x^2 - 4$, in order for its inverse to be a function.

Solution

a) $y = x^2 - 4$

$$x = y^2 - 4$$

1 mark for switching x and y

$$y^2 = x + 4$$

$$y = \pm\sqrt{x+4}$$

1 mark for solving for y

2 marks

b) Domain: $\{x \mid x \geq 0, x \in \mathbb{R}\}$

or

Domain: $[0, \infty)$

1 mark

Note:

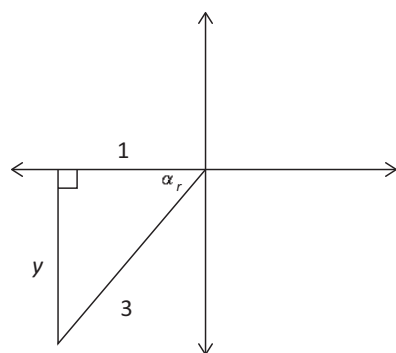
Other solutions are possible for b).

Question 31

T6

Given $\cos \alpha = -\frac{1}{3}$ and $\sin \beta = -\frac{2}{3}$ where α and β terminate in the same quadrant, determine the exact value of $\sin(\alpha - \beta)$.

Solution



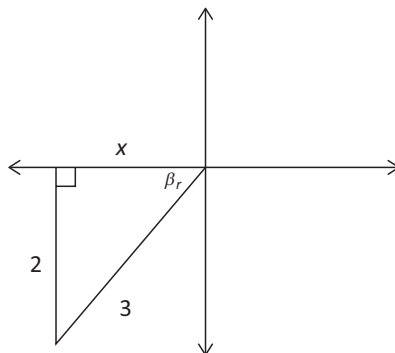
$$x^2 + y^2 = r^2$$

$$1 + y^2 = 9$$

$$y^2 = 8$$

$$y = \pm\sqrt{8}$$

$$y = -\sqrt{8}$$



$$x^2 + y^2 = r^2$$

$$x^2 + 4 = 9$$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$

$$x = -\sqrt{5}$$

$\frac{1}{2}$ mark for value of x

$\frac{1}{2}$ mark for value of y

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= \left(\frac{-\sqrt{8}}{3} \right) \left(\frac{-\sqrt{5}}{3} \right) - \left(\frac{-1}{3} \right) \left(\frac{-2}{3} \right)$$

$$= \frac{\sqrt{40}}{9} - \frac{2}{9}$$

$$= \frac{\sqrt{40} - 2}{9}$$

or

$$= \frac{2\sqrt{10} - 2}{9}$$

$\frac{1}{2}$ mark for consistent value of $\sin \alpha$

$\frac{1}{2}$ mark for consistent value of $\cos \beta$

1 mark for substitution into correct identity

3 marks

Note:

Accept any of the following values for x : $x = \pm\sqrt{5}$, $x = -\sqrt{5}$, $x = \sqrt{5}$.

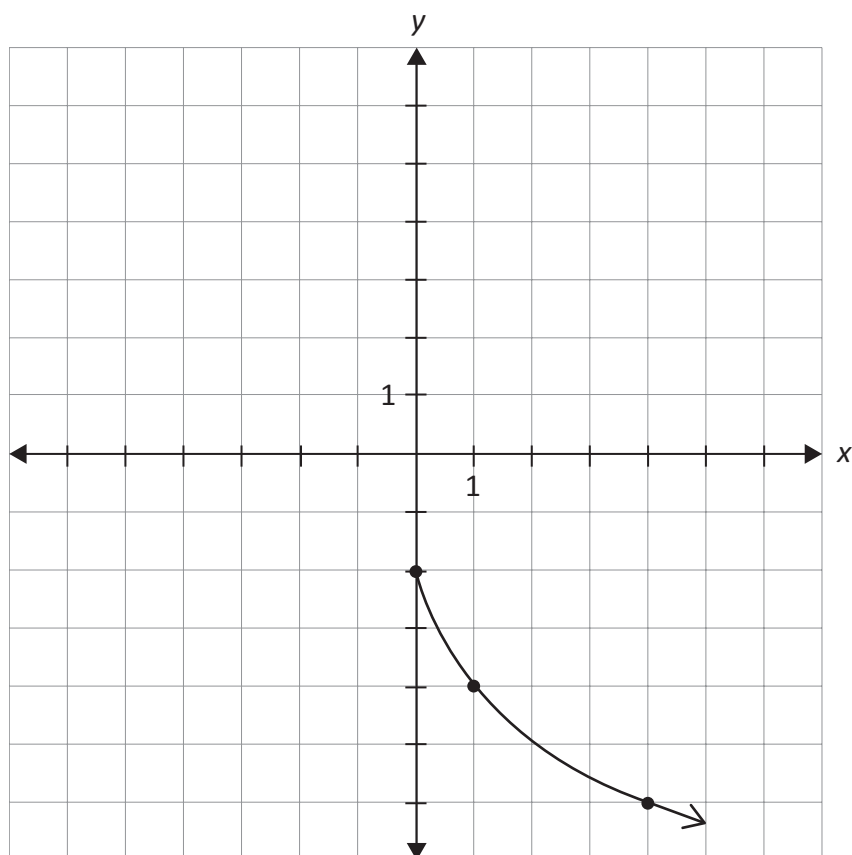
Accept any of the following values for y : $y = \pm\sqrt{8}$, $y = -\sqrt{8}$, $y = \sqrt{8}$.

Question 32

R13

Sketch the graph of $y + 2 = -\sqrt{4x}$.

Solution



- 1 mark for shape of a radical function
- 1 mark for vertical reflection
- 1 mark for horizontal compression
- 1 mark for vertical translation

4 marks

Question 33

R11

Express $p(x) = x^4 - x^3 - 6x^2 + 4x + 8$ in completely factored form, given that $(x + 1)$ is a factor of $p(x)$.

Solution

$$\begin{array}{r|rrrrr} -1 & 1 & -1 & -6 & 4 & 8 \\ & & -1 & 2 & 4 & -8 \\ \hline & 1 & -2 & -4 & 8 & 0 \end{array}$$

½ mark for synthetic division
(or equivalent strategy)

$$p(x) = (x + 1)(x^3 - 2x^2 - 4x + 8)$$

$$p(2) = (2 + 1)(2^3 - 2(2)^2 - 4(2) + 8)$$

$$p(2) = 0$$

1 mark for identifying a second zero
of $p(x)$

∴ $(x - 2)$ is a factor

$$\begin{array}{r|rrrr} 2 & 1 & -2 & -4 & 8 \\ & & 2 & 0 & -8 \\ \hline & 1 & 0 & -4 & 0 \end{array}$$

½ mark for synthetic division
(or equivalent strategy)

$$p(x) = (x + 1)(x - 2)(x^2 - 4)$$

$$p(x) = (x + 1)(x - 2)^2(x + 2)$$

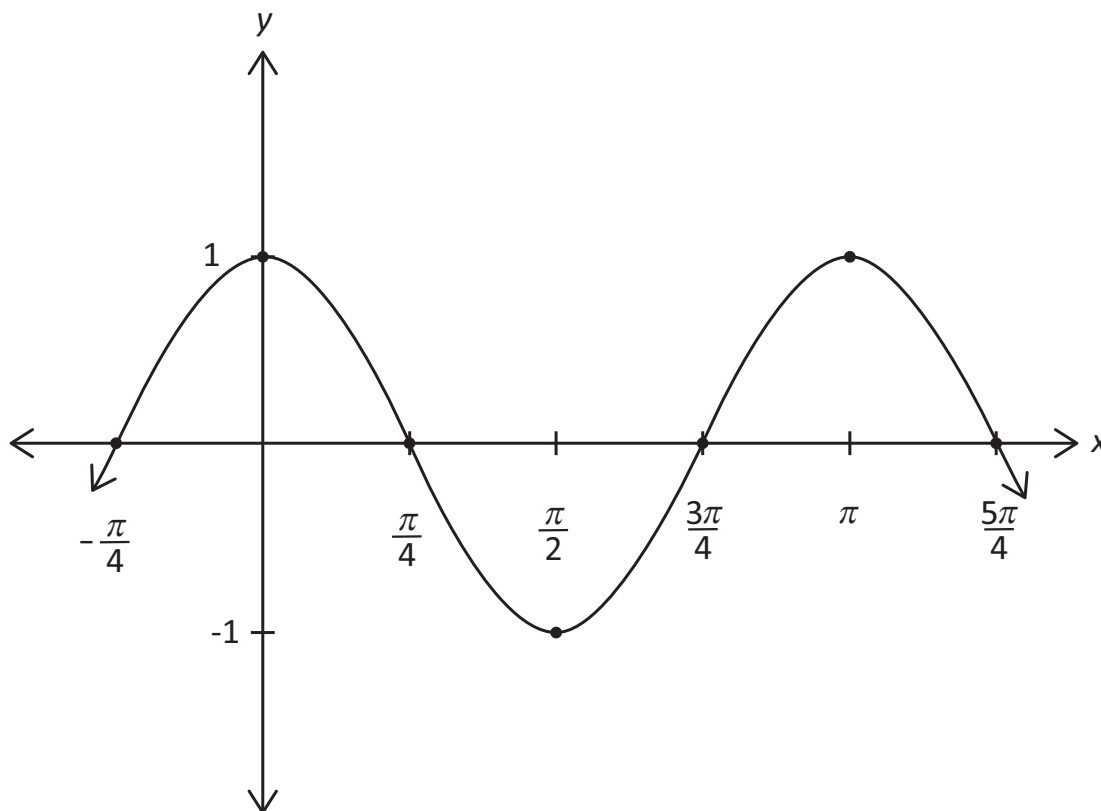
1 mark for consistent product of factors

3 marks

Question 34**T4**

Sketch at least one period of the graph of $y = -\sin\left(2\left(x - \frac{\pi}{4}\right)\right)$.

Solution



1 mark for shape of a sinusoidal function with correct amplitude

1 mark for vertical reflection

1 mark for horizontal translation

1 mark for period

4 marks

Question 35

R9

Given $g(x) = \log_2(x) - 3$,

a) determine the x-intercept of the graph.

b) sketch the graph of $g(x)$.

Solution

a) $0 = \log_2(x) - 3$

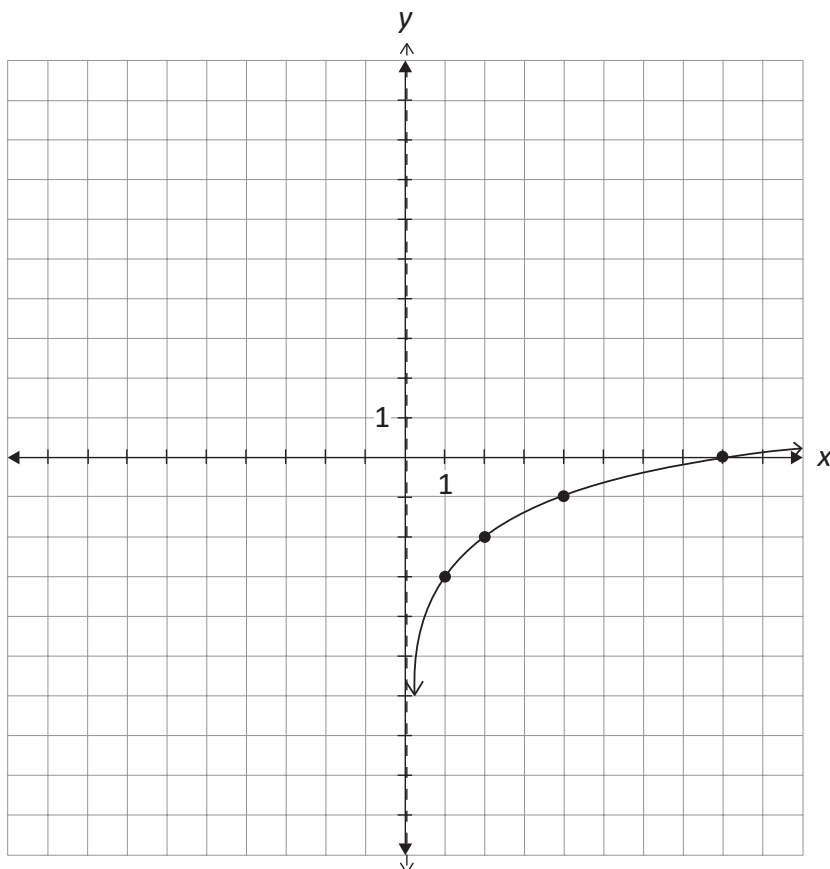
$$3 = \log_2(x)$$

$$2^3 = x$$

$$x = 8$$

1 mark

b)



1 mark for shape of increasing logarithmic function with base 2

1 mark for vertical translation

2 marks

Question 36

T6

Modesola incorrectly evaluated the expression, $\cos^2(30^\circ) - \sin^2(30^\circ)$.

Modesola's answer:

$$\begin{aligned}\cos^2(30^\circ) - \sin^2(30^\circ) \\&= 2\cos(30^\circ) \\&= 2\left(\frac{\sqrt{3}}{2}\right) \\&= \sqrt{3}\end{aligned}$$

Describe their error.

Solution

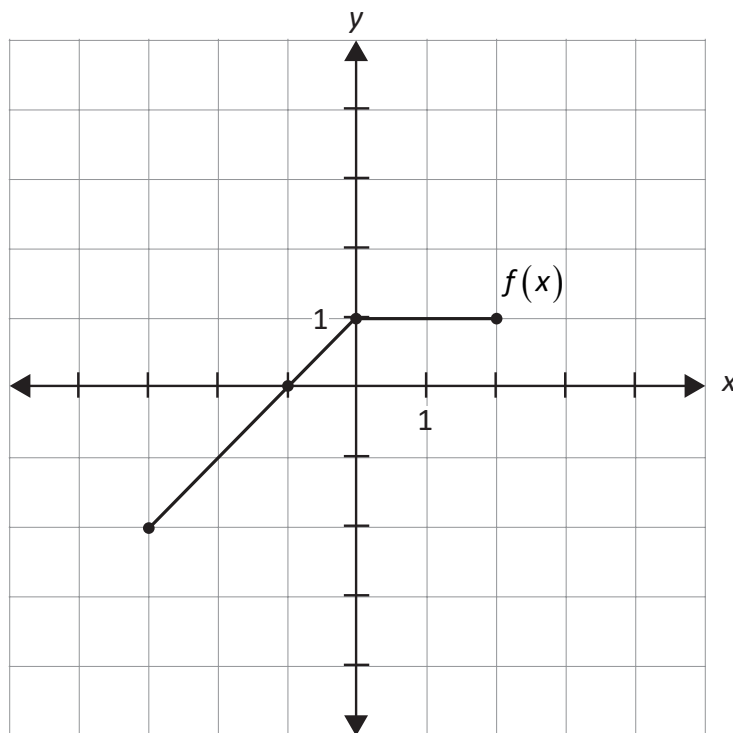
Modesola incorrectly applied the double angle identity. They should have written $\cos 2(30^\circ)$ instead of $2\cos(30^\circ)$.

1 mark

Question 37

R13

Given the graph of $y = f(x)$, state the range of $y = \sqrt{f(x)}$.



Solution

$$\{y \mid 0 \leq y \leq 1, y \in \mathbb{R}\}$$

or

$$[0, 1]$$

1 mark

Question 38

T2

Determine a value of x that satisfies $\sin^2(2x-1) + \cos^2(x+5) = 1$.

Solution

$$2x - 1 = x + 5$$

$$x = 6$$

1 mark

Question 39

R10

Solve, algebraically.

$$9^{2x+3} = \left(\frac{1}{3}\right)^{x+1}$$

Solution

$$(3^2)^{2x+3} = (3^{-1})^{x+1}$$

1 mark for changing to a common base

$$3^{4x+6} = 3^{-x-1}$$

$$4x + 6 = -x - 1$$

1 mark for equating exponents with a common base

$$5x = -7$$

$$x = -\frac{7}{5}$$

2 marks

Question 40

P2

Determine the total number of arrangements of the letters in the word GUIDES if the vowels and consonants must alternate.

Solution

Method 1

$$\underline{6} \cdot \underline{3} \cdot \underline{2} \cdot \underline{2} \cdot \underline{1} \cdot \underline{1} = 72$$

1 mark for beginning with any letter

1 mark for alternating consonants and vowels

2 marks

Method 2

$$\text{case 1: } \frac{3}{v} \cdot \frac{3}{c} \cdot \frac{2}{v} \cdot \frac{2}{c} \cdot \frac{1}{v} \cdot \frac{1}{c} = 36$$

1 mark for alternating consonants and vowels

$$\text{case 2: } \frac{3}{c} \cdot \frac{3}{v} \cdot \frac{2}{c} \cdot \frac{2}{v} \cdot \frac{1}{c} \cdot \frac{1}{v} = 36$$

½ mark for second case

$$36 + 36 = 72 \quad \frac{1}{2} \text{ mark for addition of cases}$$

2 marks

Question 41

P2

Solve, algebraically.

$$\frac{5!(n-5)!}{3!(n-6)!} = 80$$

Solution

$$\frac{5 \cdot 4 \cdot \cancel{3!} (n-5) \cancel{(n-6)!}}{\cancel{3!} \cancel{(n-6)!}} = 80$$

1 mark for factorial expansion ($\frac{1}{2}$ mark for each)
 $\frac{1}{2}$ mark for simplification of factorials

$$20(n-5) = 80$$

$$n-5 = 4$$

$$n = 9$$

$\frac{1}{2}$ mark for solving for n

2 marks

Question 42

R7

Justify that 2.1 is a better estimate than 2.6 for the value of $\log_5 29$.

Solution

$$5^2 = 25$$

$$5^3 = 125$$

$$\log_5 25 = 2$$

$$\log_5 125 = 3$$

29 is closer to 25 than 125

$\therefore \log_5 29$ is closer to 2 than 3

1 mark

Question 43

R11

When $p(x) = x^3 + kx + 6$ is divided by $(x + 2)$, the remainder is 4. Determine the value of k .

Solution

Method 1

$$4 = (-2)^3 + k(-2) + 6$$

$\frac{1}{2}$ mark for $x = -2$

1 mark for remainder theorem

$$4 = -8 - 2k + 6$$

$$4 = -2 - 2k$$

$$6 = -2k$$

$$k = -3$$

$\frac{1}{2}$ mark for solving for k

2 marks

Method 2

$$x = -2$$

$\frac{1}{2}$ mark for $x = -2$

-2		1	0	k	6
			-2	4	$-2k - 8$
		1	-2	$k + 4$	$-2k - 2$

1 mark for synthetic division (or equivalent strategy)

$$-2k - 2 = 4$$

$$-2k = 6$$

$$k = -3$$

$\frac{1}{2}$ mark for solving for k

2 marks

Question 44

R2, R3, R5

Describe the transformations used to obtain the graph of the function $y = -f(7x + 14)$ from the graph of $y = f(x)$.

Solution

- reflection over the x -axis
- horizontal stretch by a factor of $\frac{1}{7}$
- horizontal translation 2 units to the left

1 mark for vertical reflection

1 mark for horizontal stretch

1 mark for horizontal translation

3 marks

Note:

Deduct a maximum of 1 mark for concept error of incorrect order of horizontal transformations.